

ORIGINAL ARTICLE

Peripheral immune checkpoint gene expression as diagnostic biomarkers in idiopathic pulmonary fibrosis

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ABSTRACT

Background and aim: Idiopathic pulmonary fibrosis (IPF) is a relatively rare and progressive fibrotic interstitial lung disease with limited annual incidence and poor prognosis. Increasing evidence suggests that immune dysregulation, including altered immune checkpoint and innate regulatory signaling pathways, may contribute to disease pathogenesis. This study aimed to investigate peripheral blood expression levels of *PD-1*, *PD-L1*, *TIM-3*, and *TOLLIP* genes and to evaluate their diagnostic potential in patients with IPF.

Methods: This case–control study included 20 IPF patients and 20 age-matched healthy controls. Gene expression levels were quantified using real-time quantitative polymerase chain reaction (RT-qPCR) and calculated using the ΔCt method. Diagnostic discrimination was assessed by receiver operating characteristic (ROC) analysis.

Results: *PD-L1* ($p < 0.001$) and *TOLLIP* ($p = 0.002$) ΔCt values were significantly higher, indicating reduced relative gene expression. ROC analysis demonstrated strong diagnostic performance for *PD-L1* (AUC = 0.883) and *TOLLIP* (AUC = 0.865). The combined *PD-L1* + *TOLLIP* biomarker panel demonstrated superior diagnostic accuracy (AUC = 0.948). Internal validation using bootstrap resampling confirmed model stability (mean AUC = 0.950; 95% CI: 0.85–1.00).

Conclusions: In conclusion, peripheral blood *PD-L1* and *TOLLIP* expression levels are significantly downregulated in patients with idiopathic pulmonary fibrosis. The combined *PD-L1* + *TOLLIP* biomarker panel demonstrated strong diagnostic discrimination (AUC = 0.948), supporting the potential clinical utility of peripheral immune regulatory gene signatures as minimally invasive biomarkers in IPF. Larger prospective studies are warranted to validate these findings and to explore their potential role in clinical decision-making.

Key words: biomarker, IPF, PD-1, PD-L1, TIM-3, TOLLIP



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Introduction

Idiopathic pulmonary fibrosis (IPF) is a rare, chronic, and progressive interstitial lung disease characterized by progressive fibrotic remodeling of the lung parenchyma and progressive decline in pulmonary function. Despite advances in diagnostic approaches and antifibrotic therapies, IPF remains associated with poor clinical outcomes and limited treatment options. According to the most recent international clinical practice guidelines of the American Thoracic Society, European Respiratory Society, Japanese Respiratory Society, and Latin American Thoracic Association (ATS/ERS/JRS/ALAT), IPF represents one of the most severe forms of fibrotic interstitial lung disease, with a median survival of approximately three to five years following diagnosis (1). Epidemiological studies estimate the annual incidence of IPF to range between approximately 3 and 9 cases per 100,000 individuals in Europe and North America (2,3).

The pathogenesis of IPF is thought to involve repeated alveolar epithelial injury followed by dysregulated repair mechanisms that promote fibroblast activation and excessive extracellular matrix deposition. Increasing evidence suggests that interactions between injured epithelial cells, macrophages, fibroblasts, and immune regulatory signaling pathways contribute to the establishment of a profibrotic microenvironment. This complex interplay between immune signaling and fibrotic remodeling, often described as the immune–fibrosis axis, is increasingly recognized as a key driver of disease progression (4,5).

In addition to local tissue processes, systemic immune dysregulation may also contribute to disease progression. Alterations in circulating immune cells and immunoregulatory signaling pathways may reflect molecular mechanisms associated with fibrotic lung disease. Consequently, peripheral blood–based gene expression profiling has emerged as a promising non-invasive approach for identifying molecular biomarkers that may improve understanding of disease mechanisms and support clinical monitoring (6).

Among immune regulatory pathways, immune checkpoint molecules and innate immune signaling regulators have gained increasing attention in fibrotic lung

diseases. Programmed cell death-1 (*PD-1*) and its ligand programmed cell death ligand 1 (*PD-L1*) regulate immune responses and may influence fibroblast activation and myofibroblast differentiation through signaling pathways involving *SMAD3* and β -catenin (7–9). Similarly, the immune regulatory receptor T-cell immunoglobulin and mucin-domain-containing protein-3 (*TIM-3*) has been implicated in macrophage polarization and inflammatory signaling during lung injury and repair (10,11). Toll-interacting protein (*TOLLIP*), a key negative regulator of Toll-like receptor–mediated signaling, also plays an important role in maintaining epithelial homeostasis and modulating inflammatory responses associated with fibrotic remodeling (12,13).

However, despite increasing recognition of immune regulatory pathways in fibrotic lung disease, the systemic expression patterns of key immune checkpoint-related genes in peripheral blood remain insufficiently characterized in patients with idiopathic pulmonary fibrosis.

Therefore, the present study aimed to evaluate the peripheral blood expression levels of immune checkpoint and innate immune regulatory genes, including *PD-1*, *PD-L1*, *TIM-3*, and *TOLLIP*, in patients with idiopathic pulmonary fibrosis in order to characterize systemic immune dysregulation and explore their potential utility as non-invasive molecular biomarkers.

Methods

Study design and participants

This observational case–control study investigated peripheral immune checkpoint–related gene expression in patients with idiopathic pulmonary fibrosis (IPF) and healthy controls. The overall study workflow is illustrated in Figure 1.

A total of 40 participants were included, comprising 20 patients diagnosed with idiopathic pulmonary fibrosis (IPF) and 20 healthy controls. Peripheral blood samples were collected from all participants, followed by RNA isolation and complementary DNA (cDNA) synthesis. Gene expression levels of *PD-1*, *PD-L1*, *TIM-3*, and *TOLLIP* were quantified using RT-qPCR. Statistical

analyses included expression comparison between groups and receiver operating characteristic (ROC) curve analysis to evaluate diagnostic discrimination.

Inclusion and exclusion criteria for study cohort

Patients were eligible for inclusion if they were ≥ 18 years of age and had a multidisciplinary diagnosis of idiopathic pulmonary fibrosis established by the Interstitial Lung Disease Board of the Department of Chest Diseases based on clinical evaluation, high-resolution computed tomography (HRCT) findings, and when available histopathological assessment, in

accordance with the 2022 ATS/ERS/JRS/ALAT (1) clinical practice guideline.

Patients were excluded if they had active malignancy, autoimmune or connective tissue disease, other known causes of interstitial lung disease, or any comorbid condition that could significantly influence peripheral immune-related gene expression. Participants with acute respiratory infection at the time of sampling were also excluded.

To minimise potential confounding effects on immune-related gene expression, patients with conditions known to alter systemic immune responses significantly were excluded from the study. These conditions included active malignancy, autoimmune or connective tissue diseases, other interstitial lung diseases, and acute infections. Due to the exploratory nature of the design and the limited sample size, it was not possible to perform detailed adjustment for all comorbidities and concomitant medications; however, these factors were considered during patient selection and interpretation of results.

Healthy controls were age- and sex-matched individuals without chronic lung disease, fibrotic disease, autoimmune disorders, malignancy, or active infection. Controls were non-smokers and provided written informed consent before participation.

RNA isolation and cDNA synthesis

Peripheral venous blood (3 mL) was collected in EDTA-containing tubes from 20 patients with idiopathic pulmonary fibrosis and 20 healthy controls. Total RNA was isolated using the GeneAll RiboEx™ kit (Cat. No. 301-001, GeneAll Biotechnology, Korea) according to the manufacturer's protocol, including chloroform phase separation and column purification (Hybrid-R system, Cat. No. 305-101). RNA was eluted in RNase-free water and stored at -80°C until use. RNA concentration and purity were assessed spectrophotometrically, and samples with A260/280 ratios between 1.8 and 2.1 were considered acceptable. Complementary DNA (cDNA) was synthesized using the HyperScript™ First Strand cDNA Synthesis Kit (Cat. No. 601-005, GeneAll). Reverse transcription was performed at 55°C for

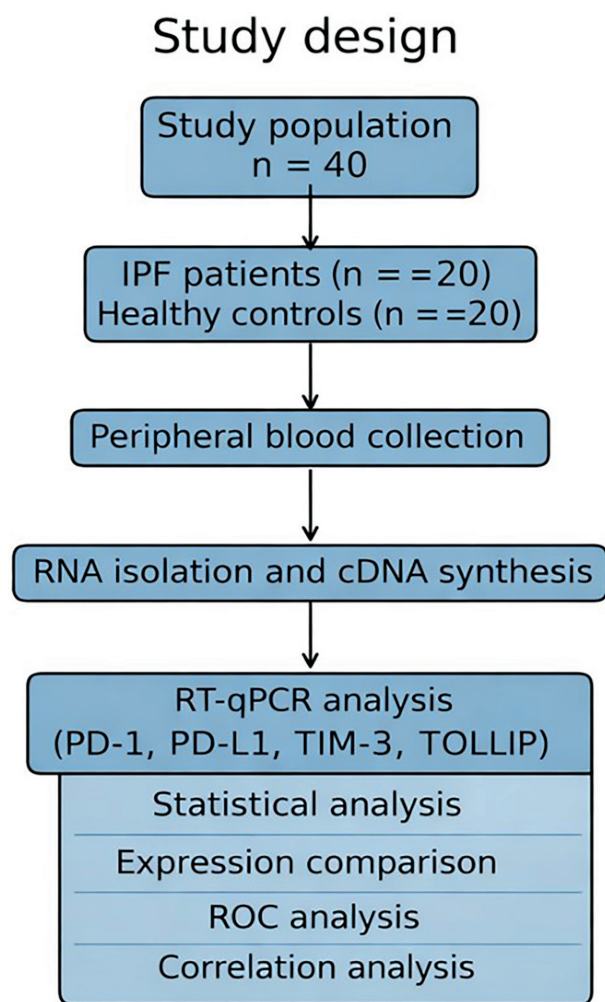


Figure 1. Study design and workflow.

60 minutes, followed by enzyme inactivation at 85°C for 5 minutes.

Quantitative real-time PCR (RT-qPCR)

The Quantitative real-time polymerase chain reaction (RT-qPCR) procedures were conducted in accordance with the Minimum Information for Publication of Quantitative Real-Time PCR Experiments (MIQE) guidelines, ensuring appropriate primer efficiency, reaction reproducibility, and amplification specificity. (RT-qPCR) was performed using A.B.T 2× SYBR Green qPCR Master Mix (Cat. No. Q03-01-05) on an Applied Biosystems™ 7500 Fast Real-Time PCR System. Reactions were carried out in a 20 µL final volume containing SYBR Green Master Mix, ROX reference dye, gene-specific primers (10 pM each) (Supplementary Table S1), cDNA template, and nuclease-free water. Thermal cycling conditions consisted of an initial denaturation at 95°C for 5 minutes, followed by 40 cycles of 95°C for 15 seconds and gene-specific annealing/extension (55–68°C) for 60 seconds. Melting curve analysis was performed to confirm amplification specificity. All samples were analyzed in triplicate, and no-template controls were included in each run. Primer amplification efficiency was considered within the acceptable MIQE range (90–110%) based on amplification curve characteristics. Gene expression levels were normalized to *GAPDH* as the endogenous reference gene, which demonstrated stable expression across study groups using ΔC_t values. For completeness, relative fold changes were additionally calculated using the $2^{-\Delta\Delta C_t}$ method and are presented in Table 2. Amplification specificity was confirmed by melt curve analysis, which demonstrated a single distinct peak for each target gene without evidence of primer-dimer formation. The overall study workflow is illustrated in Figure 1.

Statistical analysis

Statistical analyses were performed using SPSS version 21.0 (IBM Corp., USA). Normality of distribution was assessed using the Shapiro–Wilk test. Continuous variables were expressed as mean ± standard deviation or median (min–max), as appropriate.

For demographic comparisons between IPF patients and healthy controls, age was compared using Welch's t-test, while sex distribution was evaluated using Fisher's exact test. Between-group comparisons were performed using Welch's t-test when normality assumptions were satisfied and the Mann–Whitney U test when non-normal distribution was detected. A multivariable logistic regression model was constructed to evaluate the combined diagnostic discrimination of *PD-L1* and *TOLLIP* expression levels. Internal validation of the combined logistic regression model was performed using bootstrap resampling (300 iterations). Receiver operating characteristic (ROC) curves were generated to evaluate diagnostic discrimination, and area under the curve (AUC) values were calculated. Confidence intervals for AUC values were calculated using the nonparametric DeLong method. A two-sided p-value < 0.05 was considered statistically significant. For the combined biomarker analysis, a multivariable logistic regression model including *PD-L1* and *TOLLIP* ΔC_t values was constructed, and predicted probabilities derived from the model were used to generate ROC curves. Additional exploratory analyses evaluating the potential influence of demographic variables (age and sex) and treatment-related characteristics (treatment type and treatment initiation time) on gene expression levels were performed using non-parametric tests. Exploratory analyses were additionally performed to investigate potential associations between gene expression levels and clinical characteristics of IPF patients, including smoking status, environmental exposure, occupational exposure, pack-years, and forced vital capacity (FVC %) and diffusion capacity of the lung for carbon monoxide (DLCO %). Mann–Whitney U tests were used for categorical variables, and Spearman's rank correlation was used for continuous variables. Detailed statistical outputs of these analyses are provided in Supplementary Tables S2 and S3.

Ethics approval and informed consent

Ethics approval was obtained from the institutional review board (approval number: 2023/4409). Written informed consent was obtained from all participants in accordance with the Declaration of Helsinki.

Results

The study included 20 patients with idiopathic pulmonary fibrosis (IPF) and 20 age-matched healthy controls. The control group consisted of 20 healthy volunteers without known chronic lung disease, fibrotic disease, autoimmune disorders, malignancy, or active infection. Controls were non-smokers with demographic characteristics comparable to the IPF patient group. The mean age of the study population was approximately 69.6 ± 7.8 years, and controls were age- and sex-matched to the patient cohort. The overall sex distribution was 80% male and 20% female. The baseline demographic and clinical characteristics of the IPF patients are summarized in Table 1.

Table 1. Demographic and clinical characteristics of the IPF patients.

Variable	IPF Patients (n = 20)
Age (years), mean $\pm$ SD	69.7 $\pm$ 8.2
Male sex, n (%)	16 (80%)
Body mass index (kg/m ²), mean $\pm$ SD	29.98 $\pm$ 3.41
Smoking status, n (%)	
Never smokers	6 (30%)
Former smokers	14 (70%)
Pack-years, median (IQR)	30 (20–40)
Environmental exposure, n (%)	14 (70%)
Occupational exposure, n (%)	10 (50%)
Diagnosis method	
Clinical + HRCT	18 (90%)
Clinical + HRCT + biopsy	2 (10%)
Pirfenidone therapy, n (%)	15 (75%)
Nintedanib therapy, n (%)	5 (25%)
FVC (% predicted), mean $\pm$ SD	77.1 $\pm$ 14.4
DLCO (% predicted), mean $\pm$ SD	62.3 $\pm$ 15.6
Cough, n (%)	16 (80%)
Dyspnea, n (%)	16 (80%)
Digital clubbing, n (%)	14 (70%)
Velcro crackles, n (%)	18 (90%)

Abbreviations: IPF: idiopathic pulmonary fibrosis; SD: standard deviation; χ^2 : chi-square test; number: n; percentage: %.

*Table 1 summarizes the demographic and baseline clinical characteristics of idiopathic pulmonary fibrosis (IPF) patients and age-matched healthy controls. The two groups were comparable in terms of age and sex distribution, with no statistically significant differences observed ($p > 0.05$). These findings indicate appropriate matching between study groups and reduce the likelihood of confounding effects related to demographic variables. Continuous variables are presented as mean $\pm$ standard deviation (SD) where available, and categorical variables are expressed as number (n) and percentage (%).

PD-L1 gene expression

PD-L1 Δ Ct values were significantly higher in the IPF group compared with healthy controls (5.46 ± 1.20 vs. 2.16 ± 3.13). Normality testing demonstrated non-normal distribution in the control group (Shapiro–Wilk $p = 0.023$); therefore, Welch's t-test was applied. A statistically significant difference was observed between groups ($p < 0.001$), indicating reduced *PD-L1* expression in IPF patients. These findings indicate that *PD-L1* gene expression was significantly downregulated in IPF patients. Comparative gene expression data for all evaluated markers are summarized in Table 2. Corresponding fold-change values derived from $2^{-\Delta\Delta\text{CT}}$ calculations are provided in Supplementary Table S2. The distribution of *PD-L1* Δ Ct values is shown in Figure 2.

TIM-3 gene expression

TIM-3 Δ Ct values were 8.18 ± 1.61 in IPF patients and 6.66 ± 3.41 in controls. Normality testing supported parametric analysis, and Welch's t-test was performed. Although the difference was not statistically significant ($p = 0.081$), Δ Ct values tended to be higher in the IPF group.

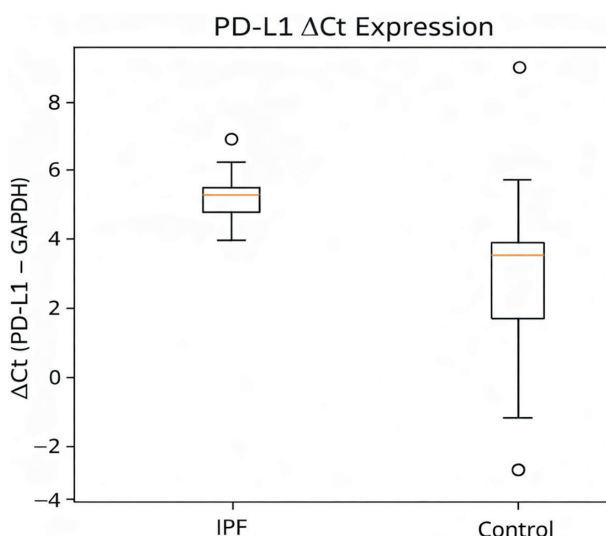
TOLLIP gene expression

TOLLIP Δ Ct values were significantly higher in IPF patients compared with controls (2.56 ± 0.50 vs. 1.67 ± 1.65). Normality assumptions were acceptable; Welch's t-test revealed a statistically significant difference between groups ($p = 0.002$), consistent with

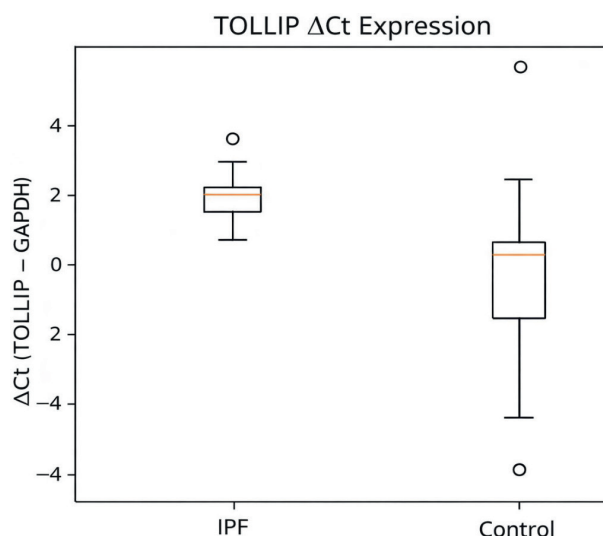
Table 2. Comparison of gene expression profiles between IPF patients and controls (ΔCt)

Gene	IPF ΔCt (mean $\pm$ SD)	Control ΔCt (mean $\pm$ SD)	Fold Change ($2^{\Delta\Delta\text{Ct}}$)	p-value	Adjusted p-value (FDR)
PD-L1	5.463 $\pm$ 1.201	2.163 $\pm$ 3.127	3.300	<0.001	<0.001
TOLLIP	2.509 $\pm$ 0.509	1.670 $\pm$ 1.650	1.623	0.002	0.004
PD-1	7.094 $\pm$ 3.048	5.299 $\pm$ 4.031	1.796	0.218	0.218
TIM-3	8.183 $\pm$ 1.605	6.660 $\pm$ 3.405	1.523	0.081	0.108

Abbreviations: ΔCt : delta cycle threshold. *Table 2 presents the comparative analysis of relative gene expression levels between IPF patients and healthy controls using ΔCt (mean $\pm$ SD) values obtained from quantitative real-time PCR analysis. Since ΔCt values are inversely proportional to gene expression, higher ΔCt values indicate lower relative expression levels. *PD-L1* and *TOLLIP* demonstrated statistically significant differences between groups, whereas *PD-1* and *TIM-3* did not reach statistical significance. P-values were adjusted for multiple comparisons using the Benjamini-Hochberg false discovery rate (FDR) method. Exact p-values were used for FDR calculations, although values are presented as $p < 0.001$ for clarity.

**Figure 2.** Peripheral blood *PD-L1* ΔCt expression levels in IPF patients and healthy controls.

Peripheral blood *PD-L1* ΔCt values were significantly higher in IPF patients compared with controls, indicating reduced relative gene expression in the IPF group. **Abbreviations:** IPF: idiopathic pulmonary fibrosis; ΔCt : delta cycle threshold.

**Figure 3.** Peripheral blood *TOLLIP* ΔCt expression levels in IPF patients and healthy controls.

Peripheral blood *TOLLIP* ΔCt values were significantly higher in IPF patients compared with controls, consistent with decreased relative expression in IPF.

Abbreviations: IPF: idiopathic pulmonary fibrosis; ΔCt : delta cycle threshold.

reduced *TOLLIP* expression in IPF. These findings indicate that *TOLLIP* gene expression was significantly downregulated in IPF patients. The distribution of *TOLLIP* ΔCt values is shown in Figure 3.

PD-1 gene expression

PD-1 ΔCt values were 7.09 $\pm$ 3.05 in the IPF group and 5.30 $\pm$ 4.03 in controls. Due to non-normal distribution in the control group (Shapiro-Wilk $p = 0.012$), the Mann-Whitney U test was applied.

No statistically significant difference was observed between groups ($p = 0.218$).

Additional analyses evaluated the potential influence of demographic and treatment-related variables on gene expression levels. No significant associations were observed between age groups, antifibrotic treatment type, or treatment initiation time and the expression levels of *PD-L1*, *TOLLIP*, or *PD-1*. However, *TIM-3* expression differed significantly according to sex ($p = 0.006$). A summary of these comparisons is presented in Table 3.

Table 3. Association of demographic and treatment variables with gene expression levels in IPF patients.

Variable	PD-L1	TIM-3	TOLLIP	PD-1
Age groups	0.488	0.719	0.908	0.365
Sex	0.962	0.006	0.776	0.202
Treatment type	0.290	0.162	0.861	0.793
Treatment initiation time	0.144	0.285	0.348	0.335

Table 4. Diagnostic discrimination of individual and combined markers (ROC Analysis)

Marker	AUC	95% CI	p-value	Optimal Cut-off	Sensitivity (%)	Specificity (%)
<i>PD-L1</i>	0.883	0.763–0.967	<0.001	4.14 (Δ Ct)	85	80
<i>TOLLIP</i>	0.865	0.722–0.977	<0.001	1.82 (Δ Ct)	95	80
<i>PD-1</i>	0.615	0.436–0.786	0.218	—	—	—
<i>TIM-3</i>	0.593	0.411–0.764	0.323	—	—	—
<i>PD-L1 + TOLLIP</i>	0.948	0.85–1.00	<0.001	0.54*	85	90

Abbreviations: ROC: receiver operating characteristic; AUC: area under the curve; Δ Ct: delta cycle threshold. Optimal cut-off values were determined using the Youden index for statistically significant ROC curves. *Table 4 presents receiver operating characteristic (ROC) curve analysis conducted to evaluate the diagnostic performance of individual gene markers and their combined predictive model in distinguishing IPF patients from healthy controls. Area under the curve (AUC), 95% confidence intervals (CI), optimal cut-off values (determined by the Youden index), sensitivity, and specificity are reported. *PD-L1* and *TOLLIP* demonstrated strong discriminatory capacity, while *PD-1* and *TIM-3* did not show statistically significant diagnostic performance. The combined *PD-L1 + TOLLIP* biomarker panel logistic regression model showed excellent diagnostic accuracy with high sensitivity and specificity. Internal validation using bootstrap resampling confirmed model robustness.

*Table 3 represents the association between demographic and treatment-related variables and gene expression levels in IPF patients. Values represent p-values obtained from non-parametric statistical tests. Bold values indicate statistical significance ($p < 0.05$).

Diagnostic discrimination (ROC Analysis)

Receiver operating characteristic (ROC) analysis was performed to evaluate the diagnostic discrimination of individual gene markers (Table 3). ROC analyses were performed using Δ Ct values obtained from RT-qPCR measurements. *PD-L1* demonstrated strong diagnostic discrimination (AUC = 0.883), with an optimal Δ Ct cut-off value of 4.14, yielding 85% sensitivity and 80% specificity. *TOLLIP* also showed strong diagnostic discrimination (AUC = 0.865), with an optimal cut-off of 1.82 (95% sensitivity and 80% specificity). *PD-1* yielded a modest AUC of 0.615 (95% CI: 0.436–0.786, $p = 0.218$) and was not considered a reliable diagnostic marker due to the absence of statistically significant discriminatory

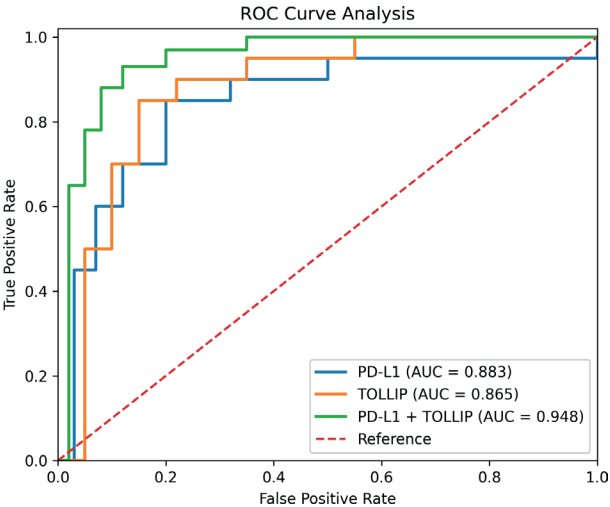


Figure 4. ROC curve of the combined *PD-L1 + TOLLIP* biomarker panel distinguishing IPF from controls.

performance. *TIM-3* also demonstrated limited diagnostic accuracy (AUC = 0.593; 95% CI: 0.411–0.764, $p = 0.323$). The ROC curve of the combined panel is presented in Table 4 and Figure 4.

Table 5. Multivariable logistic regression analysis

Variable	β Coefficient	Odds Ratio (OR)	95% CI	p-value
<i>PD-L1</i> (ΔCt)	1.390	4.013	1.555–10.359	0.004
<i>TOLLIP</i> (ΔCt)	1.397	4.045	1.287–12.709	0.017

ROC curve analysis of *PD-L1*, *TOLLIP*, and the combined logistic regression model for discrimination between idiopathic pulmonary fibrosis (IPF) patients and healthy controls. The combined *PD-L1* + *TOLLIP* panel demonstrated superior diagnostic performance (AUC = 0.948) compared to *PD-L1* alone (AUC = 0.883) and *TOLLIP* alone (AUC = 0.865). The diagonal red line represents the line of no discrimination (AUC = 0.50). The combined panel showed an improved balance of sensitivity and specificity, indicating enhanced diagnostic accuracy for IPF classification.

*ROC, receiver operating characteristic; AUC, area under the curve; IPF, idiopathic pulmonary fibrosis.

Combined *PD-L1* and *TOLLIP* Panel

To evaluate the integrated diagnostic discrimination of immune-related gene expression markers, a logistic regression model incorporating *PD-L1* and *TOLLIP* ΔCt values was constructed to assess the combined diagnostic performance (Table 5). The combined model demonstrated superior diagnostic discrimination compared with individual markers (AUC = 0.948). Using an optimal predicted probability cut-off value of 0.54 (derived from the logistic regression model), the combined biomarker panel demonstrated 85% sensitivity and 90% specificity. Internal validation using bootstrap resampling (300 iterations) confirmed model stability, yielding a mean AUC of 0.950 with a 95% confidence interval of 0.85–1.00.

These findings indicate that the combined assessment of immune checkpoint-related and innate immune regulatory gene expression enhances diagnostic accuracy beyond single-gene analysis.

* Increasing ΔCt values (reflecting lower gene expression) were independently associated with higher odds of IPF. ** Table 5 presents the results of multivariable logistic regression analysis assessing the independent association between gene expression markers and IPF diagnosis. β coefficients, odds ratios (OR),

95% confidence intervals, and p-values are shown. Increasing ΔCt values (indicating lower relative gene expression) were independently associated with higher odds of IPF for both *PD-L1* and *TOLLIP*. Panel calibration assessed by the Hosmer–Lemeshow goodness-of-fit test demonstrated adequate model fit.

Associations between gene expression levels and clinical characteristics

Exploratory analyses were performed to evaluate potential associations between peripheral gene expression levels and clinical characteristics of IPF patients. *TIM-3* expression differed significantly according to smoking status ($p = 0.012$). *PD-1* expression showed a significant association with occupational exposure ($p = 0.006$). Exploratory analyses showed no significant associations between gene expression levels and FVC. However, *TIM-3* expression demonstrated a modest positive correlation with DLCO (Spearman's $\rho = 0.449$, $p = 0.047$). No significant associations were observed between *PD-L1*, *TOLLIP*, or *PD-1* expression and DLCO (all $p > 0.05$). No significant associations were observed between gene expression levels and environmental exposure or pack-years (all $p > 0.05$) (Supplementary Table S3).

Discussion

The present study demonstrates that peripheral blood expression levels of *PD-L1* and *TOLLIP* are significantly reduced in patients with idiopathic pulmonary fibrosis, indicating alterations in systemic immune regulatory pathways associated with the disease. In contrast, *PD-1* and *TIM-3* expression levels were not significantly different between patients and healthy controls. Importantly, the combined *PD-L1* + *TOLLIP* biomarker panel demonstrated strong diagnostic discrimination with an area under the curve

(AUC) of 0.948, suggesting that integrated immune regulatory gene signatures may improve the molecular identification of IPF. Exploratory analyses further indicated that *TIM-3* expression was associated with smoking status and showed a modest correlation with DLCO, whereas *PD-1* expression was associated with occupational exposure. No associations were observed with pack-years, environmental exposure, or FVC. Moreover, the gene expression alterations identified in this study appeared largely independent of demographic characteristics and treatment-related variables, supporting the robustness of the observed findings.

While antifibrotic treatment variables were not significantly associated with gene expression levels across all four evaluated genes, only two of these genes (*PD-L1* and *TOLLIP*) demonstrated significant differences between IPF patients and healthy controls. This selective pattern of alteration further supports a disease-related signal rather than a generalised treatment effect.

Recent advances in pulmonary fibrosis research suggest that transcriptional alterations detected in peripheral blood may reflect dynamic epigenetic and regulatory mechanisms rather than fixed germline genetic variation. In idiopathic pulmonary fibrosis, environmental exposures, aging, repeated epithelial injury, and chronic inflammatory signaling are increasingly recognized as important drivers of disease progression through epigenetic modulation of immune and fibrotic pathways. Environmental factors such as cigarette smoke, air pollution, viral infections, and persistent epithelial injury can influence transcriptional regulatory networks involved in fibrotic lung disease (12).

Consequently, alterations in circulating gene expression profiles may represent systemic molecular signatures reflecting immune dysregulation rather than inherited genetic predisposition. The blood expression changes identified in this study are primarily thought to reflect systemic immune dysregulation associated with IPF, rather than pulmonary fibrotic activity. However, *PD-L1* and *TOLLIP* have been implicated in immune and fibroblast pathways, so these alterations may indirectly impact pulmonary fibrotic processes. Direct correlation between peripheral and lung tissue expression was not investigated. This distinction is important when interpreting peripheral

biomarkers in compartmentalized diseases such as IPF. Within this framework, peripheral blood gene-expression profiling may provide a noninvasive window into systemic molecular alterations associated with fibrotic lung disease and may contribute not only to biomarker discovery but also to future translational and preclinical therapeutic research.

Increasing evidence suggests that IPF progression is not solely driven by fibroblast activation but also involves dysregulated immune signaling and epithelial-immune interactions within the lung microenvironment. Persistent immune activation may amplify profibrotic signaling cascades that promote extracellular matrix deposition and progressive tissue remodeling (5,14). These mechanisms highlight the role of immune-mediated processes in shaping the fibrotic microenvironment characteristic of IPF.

The *PD-1/PD-L1* immune checkpoint pathway has gained increasing attention for its potential role in fibrotic lung diseases beyond its classical function in immune tolerance. Experimental studies have demonstrated that *PD-L1* signaling may contribute to fibroblast activation and myofibroblast differentiation through profibrotic pathways involving Smad3 and β -catenin signaling (8,9). Several studies have also reported increased *PD-L1* expression in fibrotic lung tissue and fibroblast populations in patients with idiopathic pulmonary fibrosis, suggesting that immune checkpoint signaling may participate in local tissue remodeling processes (15,16). However, these findings largely derive from lung tissue-based analyses. Emerging transcriptomic studies have demonstrated considerable cellular and molecular heterogeneity within the IPF lung microenvironment (17) and broader analyses of the *PD-1/PD-L1* pathway indicate that immune checkpoint signaling may vary depending on tissue compartment, cell type, and disease stage (18). In this context, the reduced peripheral blood *PD-L1* gene expression observed in the present study may reflect systemic transcriptional regulation of immune checkpoint pathways rather than the local protein-level alterations described within fibrotic lung tissue. These findings suggest that *PD-L1* dysregulation in IPF may be compartment-specific, with peripheral blood transcriptional signatures representing systemic immune regulatory alterations associated with the disease.

In contrast to the heterogeneous findings reported for *PD-L1*, studies investigating *TOLLIP* in pulmonary fibrosis have more consistently suggested a protective role for this molecule. *TOLLIP* is a critical negative regulator of Toll-like receptor-mediated signaling and plays an important role in maintaining epithelial homeostasis, limiting inflammatory amplification, and regulating oxidative stress responses. Previous studies have demonstrated that reduced *TOLLIP* activity may increase epithelial susceptibility to oxidative stress, mitochondrial dysfunction, and apoptosis, thereby facilitating inflammatory signaling and fibrotic remodeling processes (13,19). Genetic studies further suggest that *TOLLIP*-related pathways may influence disease susceptibility and treatment response in IPF (20,21). Within this framework, the reduced peripheral blood *TOLLIP* expression observed in our cohort may represent a systemic correlate of impaired innate immune regulation. These findings extend previous tissue-based observations and suggest that diminished *TOLLIP* activity may contribute to dysregulated inflammatory signaling and epithelial injury not only within the fibrotic lung microenvironment but also at the systemic level.

In contrast to *PD-L1* and *TOLLIP*, *PD-1* and *TIM-3* expression levels were not significantly altered in peripheral blood in our cohort. Previous investigations suggest that immune checkpoint dysregulation in IPF may occur predominantly within the pulmonary microenvironment rather than uniformly across circulating immune cells (16). *TIM-3*, for example, has been implicated in macrophage activation and inflammatory signaling during lung injury and fibrosis (11). The absence of significant systemic alterations in *PD-1* and *TIM-3* in our study, therefore, suggests that peripheral expression of these molecules may not provide reliable standalone diagnostic markers for IPF. The observed correlation between *TIM-3* expression and DLCO, although exploratory and requiring cautious interpretation, may suggest a potential relationship between immune-regulatory signaling and gas exchange impairment in IPF.

Taken together, the concomitant downregulation of *PD-L1* and *TOLLIP* in peripheral blood may indicate disruption of two complementary regulatory

systems involved in immune homeostasis: immune checkpoint signaling and innate immune inhibitory pathways. Future studies integrating peripheral gene expression signatures with radiological, functional, and clinical parameters may further improve disease stratification and diagnostic precision in idiopathic pulmonary fibrosis. This combined reduction may reflect a broader systemic immune-fibrosis axis, in which altered transcriptional regulation of immune pathways accompanies fibrotic lung disease. From a translational perspective, this interpretation is strengthened by the excellent diagnostic performance of the combined *PD-L1* + *TOLLIP* biomarker panel. These findings suggest that peripheral immune regulatory gene expression signatures may serve as non-invasive molecular biomarkers of IPF and may also provide a framework for future mechanistic and therapeutic studies targeting immune-fibrotic interactions.

From a clinical perspective, the identification of peripheral molecular signatures associated with IPF may have important diagnostic implications. Current diagnostic strategies primarily rely on high-resolution computed tomography and multidisciplinary clinical evaluation (1), while previously reported blood-based biomarkers have largely focused on markers of epithelial injury and matrix remodelling, including Krebs von den Lungen-6 (KL-6), Matrix metalloproteinase (MMP)-7, Surfactant protein A (SP-A), Surfactant protein D (SP-D), vascular endothelial growth factor (VEGF) and periostin (21–23). These established biomarkers have primarily been associated with epithelial injury, extracellular matrix turnover, or disease severity, and several have demonstrated diagnostic or prognostic utility in prior IPF cohorts. Recent biomarker studies have further emphasised the potential value of circulating molecular panels in interstitial lung diseases (24, 25). Conversely, *PD-L1* and *TOLLIP* have been shown to offer complementary insights by reflecting immune-regulatory pathways implicated in fibroblast activation, innate immune signalling, and immune-fibrotic interactions. This has the potential to enhance the efficacy of future multi-biomarker diagnostic approaches. In this context, the diagnostic performance observed for *PD-L1* (AUC 0.883) and *TOLLIP* (AUC 0.865), and particularly the combined

PD-L1 + *TOLLIP* biomarker panel (AUC 0.948), suggests that immune-regulatory gene signatures may provide complementary diagnostic information beyond conventional biomarker classes. In contrast to well-established biomarkers, which primarily reflect epithelial injury or extracellular matrix turnover, *PD-L1* and *TOLLIP* have the capacity to additionally capture disease-relevant immune dysregulation. This distinction may be clinically relevant, as *PD-L1* has been linked to profibrotic signalling and fibroblast activation, whereas *TOLLIP* has been implicated in innate immune regulation and disease susceptibility in IPF. These observations are also biologically consistent with prior studies implicating *PD-L1* in profibrotic signalling and *TOLLIP* in immune-regulatory pathways relevant to IPF pathogenesis. While these markers have not been proposed as stand-alone diagnostic tools, they may contribute to future multi-biomarker panels that complement existing diagnostic strategies if externally validated in larger cohorts. To our knowledge, this is among the first studies evaluating a combined peripheral immune checkpoint and innate immune regulatory gene expression panel for diagnostic discrimination in IPF. Several limitations should be considered when interpreting the findings of this study. First, the sample size was relatively small and derived from a single center, which may limit the generalizability of the results. The relatively small sample size should therefore be interpreted cautiously when considering the diagnostic performance estimates. Second, the cross-sectional study design precludes causal inference regarding the relationship between gene expression alterations and disease progression. Third, gene expression was evaluated only in peripheral blood and may not fully reflect molecular processes occurring within the lung microenvironment. Although all patients were receiving standard clinical management at the time of sampling, the selective alteration observed in *PD-L1* and *TOLLIP* expression—without corresponding significant changes in *PD-1* and *TIM-3* suggests that the findings are unlikely to represent a uniform medication-related effect.

Furthermore, despite the exclusion of major confounding conditions during the study's design, the potential influence of unmeasured comorbidities and

concomitant medications cannot be fully excluded and should be considered when interpreting the findings.

Future studies should aim to validate these findings in larger multicenter cohorts and to investigate the longitudinal dynamics of peripheral immune regulatory gene expression in IPF. Integrating molecular biomarkers with clinical, radiological, and functional parameters may improve disease stratification and support the development of minimally invasive diagnostic strategies for fibrotic lung diseases.

Conclusion

In conclusion, reduced peripheral *PD-L1* and *TOLLIP* expression levels were observed in patients with idiopathic pulmonary fibrosis, indicating alterations in systemic immune regulatory pathways associated with the disease. The combined *PD-L1* + *TOLLIP* biomarker panel demonstrated strong diagnostic discrimination (AUC = 0.948), suggesting that peripheral immune regulatory gene signatures may provide a minimally invasive adjunct for IPF diagnosis. Larger prospective studies are required to validate these findings and to further evaluate their potential clinical applicability.

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Authors' Contributions: All authors contributed to study conception, design, data analysis, manuscript preparation, and manuscript revision. All authors approved the final version of the manuscript.

Declaration on the Use of AI: The authors used AI-assisted tools (ChatGPT) solely for language editing and grammar

improvement. No AI tools were used for data analysis, interpretation, or scientific content generation.

Data Availability: The datasets generated and analyzed during the current study are not publicly available due to patient privacy regulations but are available from the corresponding author upon reasonable request.

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Annex

Supplementary file

Supplementary Table S1. Primer sequences, annealing temperatures, and accession numbers.

Gene	Direction	Primer Sequence (5'–3')	Annealing Temperature (°C)	Accession Number
PD-L1 (CD274)	Forward	TGCCGACTACAAGCGAATTACTG	60	NM_014143
	Reverse	CTGCTTGTCCAGATGACTTCGG		
TOLLIP	Forward	TGGAGGACAAGTGGTACAGCCT	60	NM_019009
	Reverse	ACACTGTTGGCATCAGGACCAC		
PD-1 (PDCD1)	Forward	AAGGCGCAGATCAAAGAGAGCC	60	NM_005018
	Reverse	CAACCACCAGGGTTTGGAAGT		
TIM-3 (HAVCR2)	Forward	GACTCTAGCAGACAGTGGGATC	60	NM_032782
	Reverse	GGTGGTAAGCATCCTTGGAAGG		
GAPDH	Forward	ACAAGAGGAAGAGAGAGACCC	60	NM_002046
	Reverse	TGGTTGAGCACAGGGTACTT		

* Primer specificity was confirmed by melt curve analysis, demonstrating a single amplification peak for each target gene. Relative gene expression levels were calculated using the $2^{-\Delta\Delta C_T}$ method with *GAPDH* as the endogenous reference gene.

Supplementary Table S2. Summary of statistical analyses performed in the study.

Analysis	Variable(s)	Test	p-value
Demographic comparison	Age (IPF vs control)	Welch's t-test	0.79
	Sex distribution	Fisher's exact test	1.000
Gene expression comparison	PD-L1 (ΔC_t)	Welch's t-test	<0.001
	TOLLIP (ΔC_t)	Welch's t-test	0.002
	PD-1 (ΔC_t)	Mann-Whitney U	0.218
	TIM-3 (ΔC_t)	Welch's t-test	0.081
Diagnostic discrimination	PD-L1	ROC analysis	<0.001
	TOLLIP	ROC analysis	<0.001
	PD-1	ROC analysis	0.218
	TIM-3	ROC analysis	0.323
	PD-L1 + TOLLIP	Logistic ROC model	<0.001
Multivariable analysis	PD-L1 (ΔC_t)	Logistic regression	0.004
	TOLLIP (ΔC_t)	Logistic regression	0.017
Age-related analyses	PD-L1, TIM-3, TOLLIP, PD-1	Kruskal-Wallis	>0.05
Sex-related analyses	PD-L1, TOLLIP, PD-1	Mann-Whitney U	>0.05
	TIM-3	Mann-Whitney U	0.006
Treatment-related analyses	All genes	Mann-Whitney U	>0.05
Treatment initiation time	All genes	Kruskal-Wallis	>0.05

Supplementary Table S3. Exploratory analyses of associations between clinical characteristics and gene expression levels in IPF patients.

Clinical variable	Gene (Δ Ct)	Statistical test	p-value
Smoking status	<i>PD-L1</i>	Mann–Whitney U test	0.547
	<i>TOLLIP</i>	Mann–Whitney U test	0.779
	<i>PD-1</i>	Mann–Whitney U test	0.153
	<i>TIM-3</i>	Mann–Whitney U test	0.012
Environmental exposure	<i>PD-L1</i>	Mann–Whitney U test	0.547
	<i>TOLLIP</i>	Mann–Whitney U test	0.312
	<i>PD-1</i>	Mann–Whitney U test	0.779
	<i>TIM-3</i>	Mann–Whitney U test	0.051
Occupational exposure	<i>PD-L1</i>	Mann–Whitney U test	0.307
	<i>TOLLIP</i>	Mann–Whitney U test	0.076
	<i>PD-1</i>	Mann–Whitney U test	0.006
	<i>TIM-3</i>	Mann–Whitney U test	0.345
Pack-years	<i>PD-L1</i>	Spearman correlation	>0.05
	<i>TOLLIP</i>	Spearman correlation	>0.05
	<i>PD-1</i>	Spearman correlation	>0.05
	<i>TIM-3</i>	Spearman correlation	>0.05
FVC (% predicted)	<i>PD-L1</i>	Spearman correlation	0.467
	<i>TOLLIP</i>	Spearman correlation	0.367
	<i>PD-1</i>	Spearman correlation	0.781
	<i>TIM-3</i>	Spearman correlation	0.580
DLCO (% predicted)	<i>PD-L1</i>	Spearman correlation	0.229
	<i>TOLLIP</i>	Spearman correlation	0.691
	<i>PD-1</i>	Spearman correlation	0.667
	<i>TIM-3</i>	Spearman correlation	0.047

*Associations between gene expression levels and clinical characteristics were evaluated using Mann–Whitney U tests for categorical variables and Spearman rank correlation for continuous variables. These analyses were exploratory and were not adjusted for multiple comparisons due to the limited sample size. P-values are presented. Bold indicates statistical significance ($p < 0.05$). ** Δ Ct, delta cycle threshold.